Trend Strength Analysis

Stock Return Predictability Using OHLCV Data

Portfolio Research Project

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1 Objective

The objective of this analysis is to investigate whether the strength of a stock's recent price trend contains information about its subsequent 20-day return.

The central question is:

> Does a stock with a stronger recent trend tend to generate higher future returns?

A positive relationship would provide evidence in favor of trend continuation, while a negative relationship would suggest that stronger recent trends are followed by weaker returns, which is consistent with short- or medium-term reversal.

2 Daily Returns

The daily return for stock i on day t is calculated as

$$r_{i,t} = \frac{P_{i,t}}{P_{i,t-1}} - 1,$$

where $P_{i,t}$ represents the closing price.

These daily returns are subsequently used to calculate the rolling 20-day volatility.

3 Construction of Trend Strength

The trend strength indicator is defined using two components:

1. Historical 20-day return
2. Historical 20-day volatility

The historical 20-day return is

$$R_{i,t}^{20} = \frac{P_{i,t}}{P_{i,t-20}} - 1.$$

The 20-day rolling volatility is calculated as the standard deviation of daily returns over the previous 20 trading days:

$$\sigma_{i,t}^{20} = Std(r_{i,t-19}, \ldots, r_{i,t}).$$

Trend strength is then defined as

$$TS_{i,t} = \frac{R_{i,t}^{20}}{\sigma_{i,t}^{20}}.$$

Thus, the indicator measures the magnitude of the recent price movement relative to the volatility experienced by the stock.

A high value indicates that the stock experienced a relatively strong positive recent trend compared with its volatility.

A low value indicates either a weak recent trend or a relatively high level of volatility.

4 Future Return

To determine whether trend strength has predictive power, the subsequent 20-day return is calculated.

The future 20-day return is

$$FR_{i,t}^{20} = \frac{P_{i,t+20}}{P_{i,t}} - 1.$$

The trend strength at time t is therefore compared with the return realized over the following 20 trading days.

5 Cross-Sectional Decile Analysis

For each trading day, stocks are ranked according to their trend strength.

The stocks are divided into ten groups:

- Group 1: weakest trend strength
- Group 2: relatively weak trend strength
- $\vdots$
- Group 10: strongest trend strength

The average future 20-day return is calculated separately for each group.

The results are shown below.

Table 1: Average 20-Day Future Return by Trend Strength Group

Trend Strength Group	Average Future Return
Group 1	0.022150
Group 2	0.021307
Group 3	0.020277
Group 4	0.020839
Group 5	0.016854
Group 6	0.017742
Group 7	0.016272
Group 8	0.014440
Group 9	0.011701
Group 10	0.006940

The results show a clear decreasing pattern.

The weakest-trend group generated an average subsequent 20-day return of

$$2.2150\%.$$

In contrast, the strongest-trend group generated an average subsequent 20-day return of

$$0.6940\%.$$

Therefore, stocks with stronger recent trends tended to produce lower subsequent returns in this dataset.

6 Interpretation of the Decile Results

The decile results provide evidence against the hypothesis that stronger recent trends lead to stronger future returns.

Instead, the relationship is predominantly negative.

The difference between the weakest and strongest groups is

$$2.2150\% - 0.6940\% = 1.5210\%.$$

Thus, the extreme-group spread is

$$\boxed{1.5210\%}.$$

This pattern is consistent with a reversal effect.

In other words, stocks that had performed strongly over the previous 20-day period tended to experience weaker returns over the following 20-day period.

Similarly, stocks with weak recent trends tended to generate higher subsequent returns.

However, the decile analysis alone does not establish statistical significance. Additional statistical tests are therefore required.

7 Information Coefficient Analysis

The Information Coefficient (IC) measures the cross-sectional relationship between the trend strength signal and future returns.

For each day, the correlation between trend strength and subsequent 20-day return is calculated.

The resulting summary statistics are:

Table 2: Information Coefficient Results

Statistic	**Value**
Mean IC	-0.069224
Median IC	-0.070845
IC Standard Deviation	0.248725

The mean IC is

$$\boxed{-0.069224}.$$

The negative sign indicates that higher trend strength is associated with lower subsequent 20-day returns.

The median IC is also negative:

$$\boxed{-0.070845}.$$

The fact that both the mean and median IC are negative provides further evidence that the relationship is not driven solely by a small number of extreme observations.

The magnitude of the IC is moderate rather than extremely large. Therefore, trend strength should not be interpreted as a deterministic predictor of future returns.

8 Long–Short Portfolio Analysis

A simple long–short portfolio is constructed to examine whether the observed relationship can be translated into a portfolio-level signal.

The portfolio takes:

- Long position in the weakest-trend group

- Short position in the strongest-trend group

This construction is appropriate because the observed relationship is negative.

The resulting long–short spread is

$$2.2150\% - 0.6940\% = 1.5210\%.$$

The measured portfolio statistics are:

Table 3: Long–Short Portfolio Results

Statistic	**Value**
Mean Return	0.015210
Standard Deviation	0.067607
Sharpe-like Ratio	0.2250

The mean return of the long–short portfolio is

$$\boxed{0.015210}.$$

The standard deviation is

$$\boxed{0.067607}.$$

The resulting Sharpe-like ratio is

$$\boxed{0.2250}.$$

The positive long–short return is consistent with the negative relationship observed in the IC and decile analysis.

However, the relatively low Sharpe-like ratio indicates that the signal also experiences considerable variability.

9 Overlapping Return Problem

An important statistical issue arises because the analysis uses 20-day forward returns.

For example, the 20-day future return calculated on day t overlaps with the 20-day future return calculated on day $t + 1$.

Therefore, consecutive observations are not independent.

This creates serial correlation in the observations and can cause ordinary standard errors to underestimate the true uncertainty.

Consequently, a conventional t-test may overstate the statistical significance of the observed relationship.

To address this issue, a Newey–West heteroskedasticity and autocorrelation consistent (HAC) standard error is used.

10 Newey–West HAC Test

A Newey–West correction with 20 lags is applied to account for the overlapping 20-day return horizon.

The results are:

Table 4: Newey–West HAC Results

Statistic	**Value**
Mean Return	0.015210
HAC Standard Error	0.004996
HAC t-statistic	3.0447
HAC p-value	0.002329

The HAC t-statistic is

$$\boxed{3.0447}.$$

The corresponding p-value is

$$\boxed{0.002329}.$$

The p-value is below the conventional 5% significance level.

Therefore, after accounting for the dependence caused by overlapping 20-day returns, the observed long–short relationship remains statistically significant in this sample.

This strengthens the evidence that the negative relationship between trend strength and future returns is not simply a consequence of underestimated standard errors.

11 Problems and Limitations

Although the results are statistically significant, several limitations must be considered.

11.1 Short Sample of Stocks

The dataset contains only 24 stocks.

Therefore, the cross-sectional analysis is based on a relatively small number of securities on each day.

This can make portfolio construction and rank-based estimates less stable than they would be in a much larger universe.

11.2 Overlapping Returns

The 20-day forward return creates substantial overlap between consecutive observations.

This violates the independence assumption of a simple t-test.

The Newey–West correction reduces this problem, but it does not eliminate every possible source of dependence.

11.3 No Transaction Costs

The analysis does not explicitly include:

- Brokerage costs
- Bid–ask spreads
- Slippage
- Market impact
- Taxes or other trading costs

Therefore, the reported portfolio returns represent theoretical returns before implementation costs.

11.4 No Position Constraints

The analysis does not incorporate practical portfolio constraints such as:

- Maximum position size
- Sector constraints
- Liquidity constraints
- Short-selling restrictions
- Turnover limits

Such constraints may affect the actual performance of the strategy.

11.5 Historical Relationship

The analysis identifies a relationship within the available historical sample.

A statistically significant historical relationship does not guarantee that the same relationship will continue in the future.

Further out-of-sample testing is therefore required.

11.6 Potential Data-Snooping Risk

The trend-strength specification uses a 20-day lookback and a 20-day forward horizon.

Testing many possible combinations of lookback periods, volatility windows, and holding periods could increase the risk of finding a relationship by chance.

Consequently, the results should be interpreted within the context of the particular specification tested.

12 Comparison with Reversal

The trend-strength result is closely related to the reversal analysis performed earlier.

The reversal analysis produced a mean IC of approximately

$$-0.0673,$$

while trend strength produced

$$-0.069224.$$

Both values are negative and of similar magnitude.

This suggests that trend strength is capturing information related to the same underlying short- or medium-term reversal behavior.

However, the portfolio-level results differ.

The reversal strategy produced a higher Sharpe ratio, whereas the trend-strength strategy produced a lower Sharpe-like ratio of

$$0.2250.$$

Therefore, the trend-strength signal appears to provide supporting information about reversal rather than representing a completely independent source of predictive information.

13 Overall Interpretation

The central hypothesis was that stronger recent trends would lead to higher future returns.

The empirical results do not support this hypothesis for the dataset and time horizon examined.

Instead, the relationship is consistently negative.

The decile analysis shows that the weakest-trend stocks generated the highest subsequent returns, while the strongest-trend stocks generated the lowest subsequent returns.

The IC analysis produces a negative mean IC of

$$-0.069224,$$

and the median IC is also negative:

$$-0.070845.$$

The long–short portfolio produces a positive spread of

$$1.5210\%.$$

After accounting for overlapping observations using a Newey–West correction, the relationship remains statistically significant with

$$t = 3.0447$$

and

$$p = 0.002329.$$

Therefore, the evidence in this sample is more consistent with short- to medium-term reversal than with trend continuation.

14 Final Conclusion

The main conclusions from the trend-strength analysis are:

1. The dataset contained 35,616 observations across 24 stocks.
2. The weakest-trend group produced an average future return of 2.2150%, while the strongest-trend group produced only 0.6940%.
3. The resulting extreme-group spread was 1.5210%.
4. The mean IC was -0.069224 and the median IC was -0.070845.
5. The long–short portfolio produced a mean return of 1.5210% with a standard deviation of 6.7607%.
6. The Sharpe-like ratio was 0.2250.
7. After accounting for overlapping returns using a 20-lag Newey–West correction, the HAC t-statistic was 3.0447 with a p-value of 0.002329.

Overall, the results indicate that stronger recent trends were associated with lower subsequent 20-day returns in the sample.

Thus, trend strength does not behave as a conventional momentum signal in this analysis. Instead, it exhibits behavior consistent with short- or medium-term reversal.

The statistical evidence remains significant after correcting for overlapping observations. However, the relatively low Sharpe-like ratio and the limitations of the dataset indicate that trend strength should be treated as a supporting signal rather than as a standalone portfolio strategy.